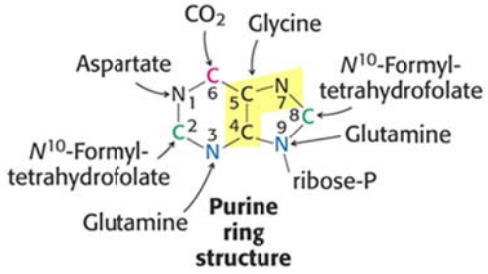
Nucleotides metabolism

- The biological processes in which nucleotides participate are numerous
- It is therefore essential that cells have a continuous supply of them
    - They are the precursors of the Ac. Nucleic DNA and RNA
    - They are the universal energy donors: ATP and GTP
    - They intervene in biosynthetic processes as activators of many molecules that must polymerize: an example is UDP-Glu
    - They are essential in the intracellular signaling cascades: cAMP, cGMP
    - They are precursors of cofactors: NAD^+, FAD, CoA, SAM

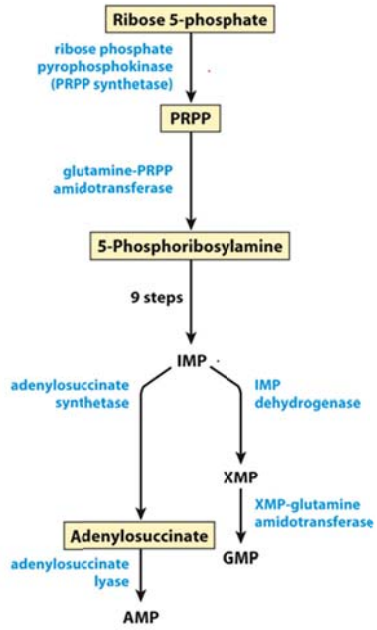
- The pathways for the nucleotide synthesis are also important as therapeutic targets for the treatment of tumors and infectious diseases.
- Cells follow two paths to synthesize nucleotides:
    - **"de novo¨ synthesis** routes: they use simple precursors to synthesize the nitrogenous bases
    - **salvage pathways**: they recycle nitrogenous bases and nucleosides derived from nucleotide degradation
- De novo pathways synthesize ribonucleotides, deoxyribonucleotides are synthesized from the corresponding ribonucleotides through a reaction that replaces the OH group of C2 in ribose by a H
- It should be noted that there are no nucleotide stores, with the exception of ATP, so cells must continually synthesize them.
- Otherwise, important processes such as DNA replication or RNA transcription may be compromised
- Both at the synthesis of purines and pyrimidines there are several precursors that are common
    - **Ribose 5P**, must be previously activated at its carbon 1 through a pyrophosphorylation reaction that gives rise to 5 'phosphoribosyl-1-pyrophosphate (PRPP)
    - **Gly, glutamine and aspartate** are amino acids that provide the nitrogen

De novo purine nucleotide synthesis

- In the case of purine nucleotides, the nitrogenous base is synthesized bound ribose.

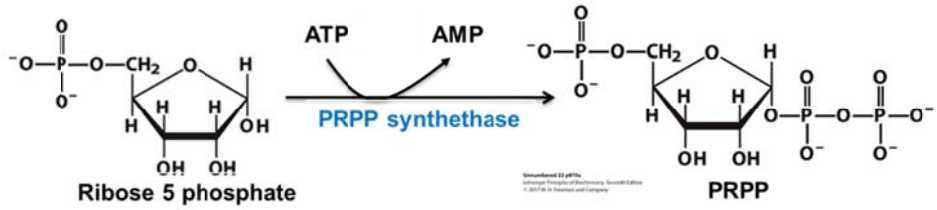


- In this image you can see the origin of each atom of the purine ring



- The purine route first generates the nucleotide called inositate (IMP) whose nitrogen base is hypoxanthine
- - From IMP the pathway follows two ways to synthesize AMP or GMP

- The first reaction, common to purines and pyrimidines, is the activation of R5P, whose primary source is the pentose phosphate pathway



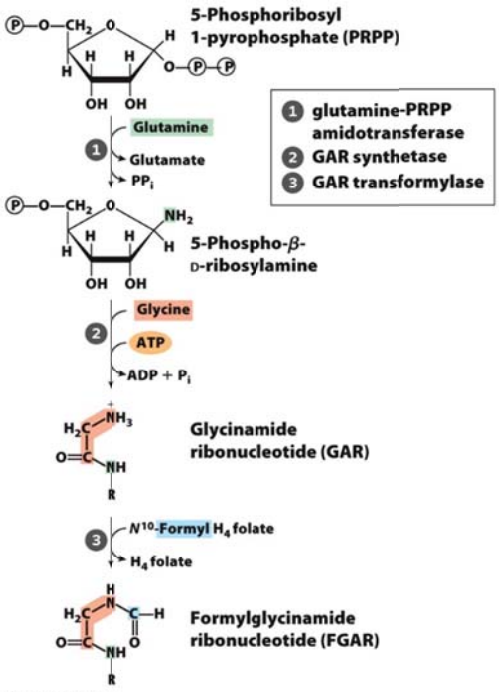
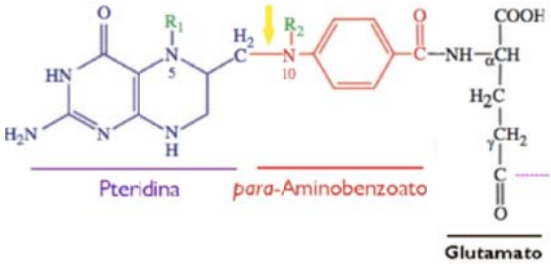


Figure 22-35 part 1
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- The reaction that commits PRPP to purine synthesis is the incorporation of an amino group from Gln onto the C1 of ribose.
- Then 3 atoms from Gly are incorporated. In this reaction, 1 ATP is hydrolyzed to promote condensation.
- The amino group of glycinamide is then formylated using formyl-tetrahydrofolate as a donor of that group



- Tetrahydrofolate is synthesized from folic acid (vit. B9). It is an important cofactor in numerous chemical reactions as a donor of formyl, methyl groups, etc. Its deficit causes anencephaly, spina bifida, premature births, megaloblastic anemia, among others

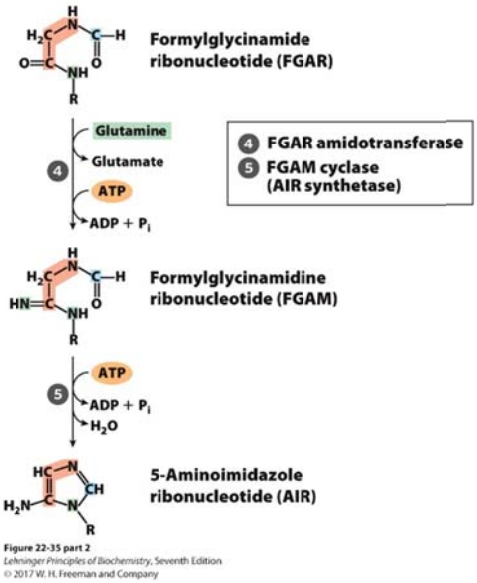


Figure 22-35 part 2
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- Then, a nitrogen atom from the amino group of Gln is incorporated

- The imidazole ring is then closed by dehydration that requires the use of ATP

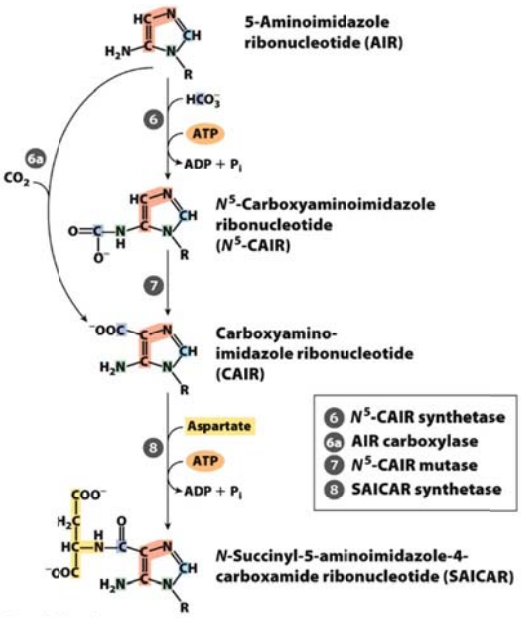


Figure 22-15 part 3
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- A carboxyl group is then incorporated by carboxylation

- The amino group of Asp is then incorporated in two steps. First an amide bond is formed between the amino of the Asp and the newly incorporated carboxyl at the previous step; then the bond between the alpha carbon and the amino is broken, releasing fumarate

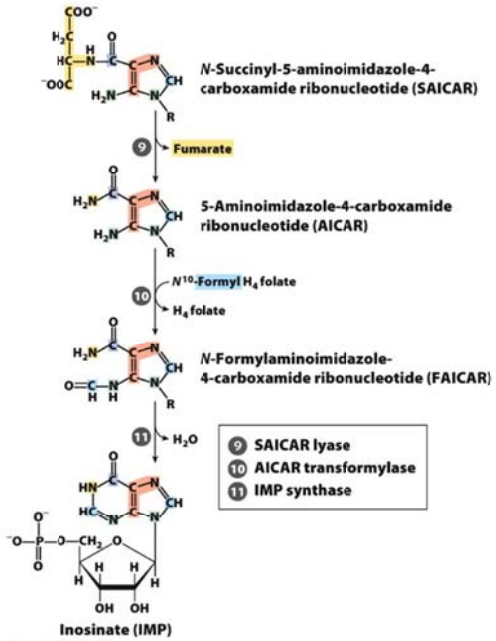


Figure 22-35 part 4
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- The last atom of the purine ring is a formyl group from formyl tetrahydrofolate again.

- Finally, the closure of the ring occurs through dehydration that gives rise to IMP.

- In humans, most of the enzymes in this pathway are found to form multienzymatic complexes.

- From this point the path forks to generate the AMP and the GMP

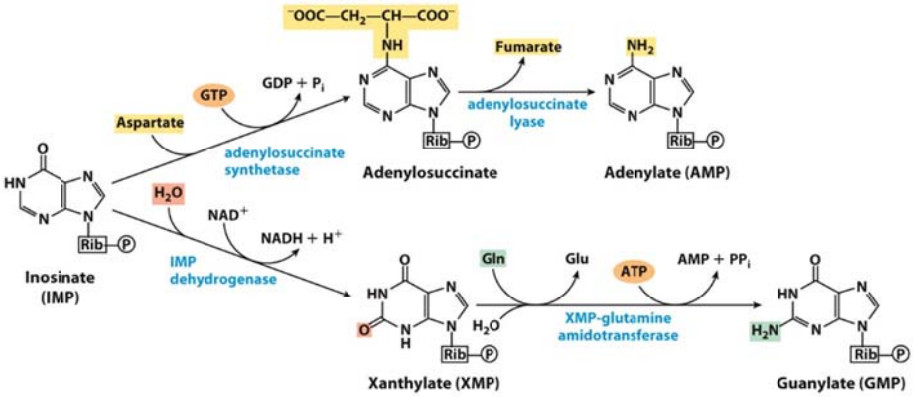


Figure 22-36
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Synthesis of AMP and GMP from IMP

- The conversion of IMP to AMP takes place through the incorporation of an amino group from Asp.
- First the Asp joins the C6 of the ring to form adenylosuccinate; in this reaction GTP is used as the energy donor
- Then the adenylosuccinate is broken, releasing fumarate and 6, aminopurine
- GMP is formed by oxidation of IMP at C2 and subsequent incorporation of an amino group from Gln.
- In this case the energy donor is ATP

Purine nucleotide biosynthesis is regulated by feedback inhibition

- Regulation of purine nucleotide synthesis offers a wonderful opportunity to review the feedback inhibition

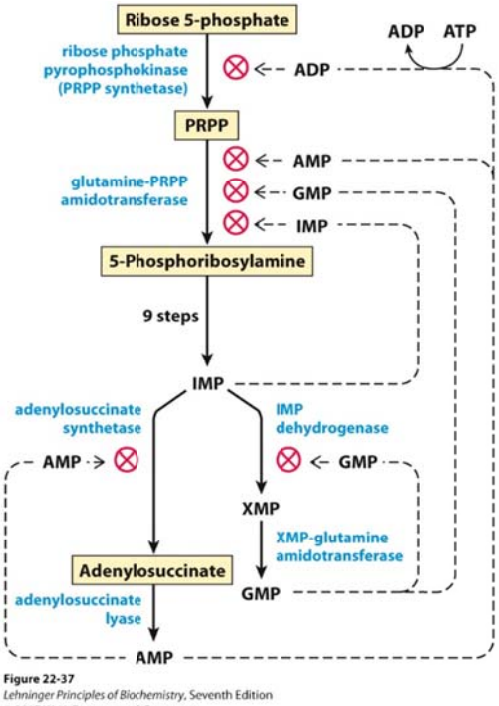


Figure 22-37
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- Regulation occurs in three levels:

- **the first level** is located at the activation reaction of ribose 5P to PRPP; This reaction requires **ATP** so the absence of it, or the increase in ADP, **inhibits the reaction**

- **the second level** is **located at the first specific reaction of purine synthesis, catalyzed by Gln-PRPP amidotransferase**.

- This enzyme is **inhibited by** the three main end products: **IMP, AMP and GMP**. IMP has a very strong inhibitory effect, while AMP or GMP exert a mild inhibition, when they act alone. This allows that if there is an excess of AMP, the pathway is not completely stopped and the

synthesis of GMP can be balanced, and vice versa.

- On the other hand, when there is an excess of both, AMP and GMP, an inhibitory synergy occurs

- the third level of regulation **is at the point where the path forks**, that is, in the conversion of IMP into AMP or GMP.

- **AMP inhibits adenylsuccinate synthetase**, favoring IMP to be used to synthesize GMP; likewise, **GMP inhibits IMP dehydrogenase**, favoring AMP synthesis.

Purines recycle

- During the nucleic acids and nucleotides degradation free nitrogenous bases are generated.
- **In the case of purine bases**, **the final product** of their degradation **is uric acid,** cleared by urine
- **A significant amount of purines**, on the other hand, **are used directly to** synthesize new purine nucleotides through salvage pathways.
- The advantage of these routes is twofold:
    - **decrease the amount of uric acid** to be excreted
    - **reduce the energy cost** of the de novo synthesis
- The biochemical mechanism of purine recovery involves the condensation of the base with activated ribose in the form of PRPP.
- In humans there are two enzymes for the salvage

- One of them **recycles adenine**: adenosine phosphoribosyl transferase

Adenine + PRPP→ AMP + PPi

- The second **recycles both hypoxanthine or guanine**: hypoxanthine-guanine phosphoribosyl transferase

Hipoxantine + PRPP→ IMP + PPi

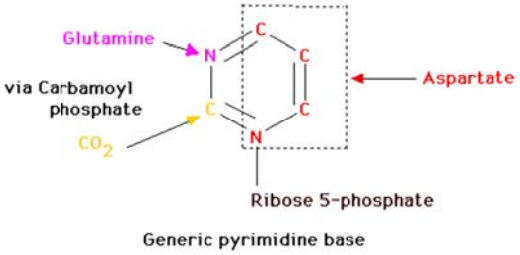
Guanine + PRPP→ GMP + PPi

- We can get an idea about the importance of purine recovery if we pay attention to the serious consequences of the decrease in the activity of the HGPRT enzyme.
- **HGPRT deficiency leads** to a syndrome called **Lesch-Nyhan**. This genetic disease affects children and causes very extreme symptoms: intellectual deficit, hostile behavior and a tendency to self-mutilation of fingers, lips, etc.
- **The deficit of the recovery capacity** causes an **extreme increase in the production of uric acid** that, in addition to gout, gives rise to a great toxicity at the CNS level.
- There is a high level of purine recovery in the fetal and postnatal brain, as well as in erythrocytes, platelets and bone marrow

- Recovery pathways are also very important for the survival of resting T lymphocytes. In addition, the activation of LT requires de novo synthesis.
- Cells infected by HIV have de novo synthesis inhibited, so when LTs are activated as a result of infection, they do not have enough nucleotides, so they die
- Parasites such as mycoplasma, chlamydia or borrelia, also take advantage of the de novo synthesis of nucleotides in the cells they parasitize

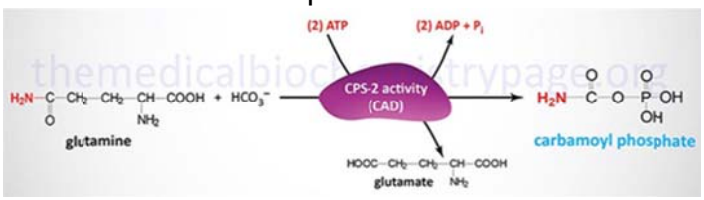
Pyrimidine synthesis

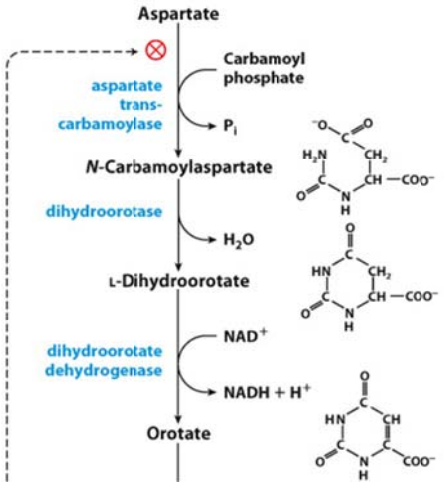
- The synthesis of pyrimidines, unlike purines, takes place through the synthesis of the nitrogenous base and then its binding to ribose
- Another difference is that this route is linear: first the UTP is synthesized and, from it, the CTP



- In the image we can observe the precursors of the purine ring atoms
- In this case the precursors are: Asp, Gln and CO_2

- The first reaction consists on the condensation of the amino group of Gln and CO_2 catalyzed by the cytoplasmic isoform of **carbamyl phosphate synthetase II**
- In this reaction 2 ATP are consumed. One to phosphorylate the carbamyl group and the other to shift the equilibrium



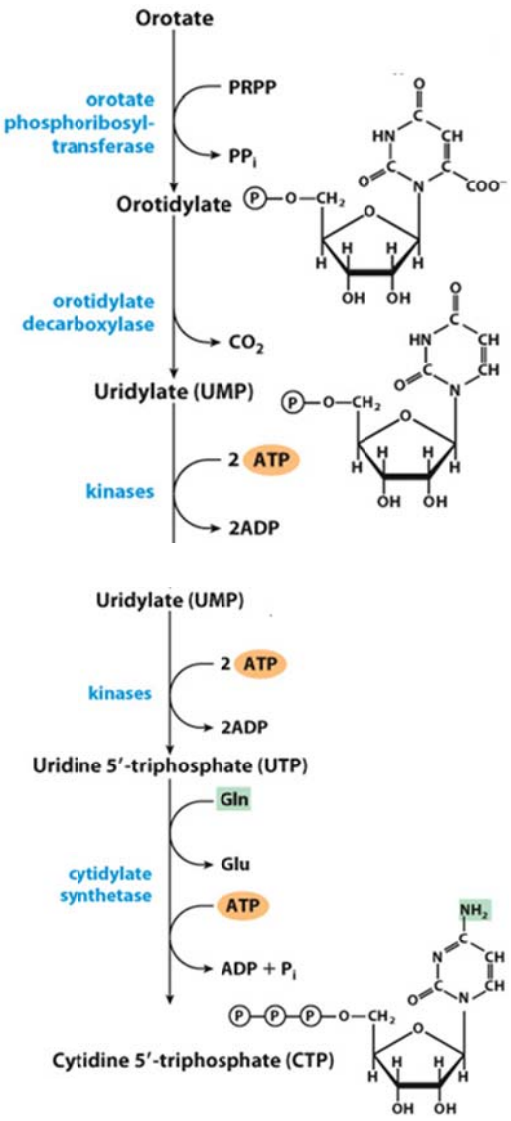


- In the next reaction, carbamyl P condenses with Asp to form carbamyl aspartate. This reaction is catalyzed by AspartateransCarbamoylase (ATC). This reaction is one of the principal control points of the route

-The ring is then closed by dehydration catalyzed by dihydrootase.

-In humans, these first three reactions are catalyzed by a multi-enzyme complex that contains the three enzymatic activities described and is called CAD because of the first letter of the name of each enzyme that is part of the complex

-Then the dihydroorotate is oxidized giving rise to the orotate



-Once the pyrimidine ring has formed, the ribose 5-phosfate binds to it. Using PRPP as a precursor and a reaction identical to that of the purine recovery pathways, this step generates a nucleotide called orotidylate

-In the next stage, the orotate is decarboxylated, and uridine is generated, with which the final product is the UMP

-UMP is phosphorylated twice to form UTP by two kinases

-Finally the UTP will be aminated to give rise to the CTP. This reaction requires Gln as the amino and ATP donor. It is catalyzed by cytidylate synthetase

-CTP levels regulate the pathway through allosteric inhibition of the CAD complex.

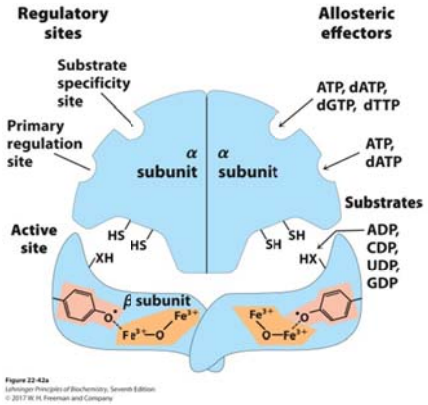
-On the other hand, ATP has an enhancing activity action since it eliminates the inhibitory effect of CTP.

-In this case, ATP is indicating that there is enough energy and that the cell is ready to replicate

-During the apoptosis process, Caspase 3 degrades the CAD complex, so the synthesis of pyrimidines is stoped

Synthesis of Deoxyribonucleotides (dNTP)

- dNTPs, necessary for DNA synthesis, are derived from their corresponding NTPs by reducing ribose C2 to a deoxy derivative.
- The reaction is catalyzed by an enzyme called **Ribonucleotide reductase** and the **substrates are ribonucleotides diphosphate**
- During the reduction process, two H atoms will be necessary, which will be provided by NADPH through an intermediate that is **thioredoxin**, a ubiquitous protein, or **glutaredoxin**.



- Ribonucleotide reductase is made up of an $\alpha_2\beta_2$ heterodimer.
- The α subunits are the catalytic ones and the β subunits are those that generate the radicals during the reaction
- The active center is at the interface between the αβ heterodimer and can accommodate any of the four nucleoside diphosphates: ADP, CDP, GDP and UDP

- The enzyme has a regulation both for its activity and for the specificity of the substrate, in this way the four dNTPs can be synthesized in an equimolecular way

- Regarding the activity, the general idea is that ATP activates the enzyme and dATP inhibits it

- Regarding specificity, the general idea is that ATP or dATP increase the affinity for CDP or UDP; while dTTP favors affinity for GDP and dGTP favors affinity for ADP

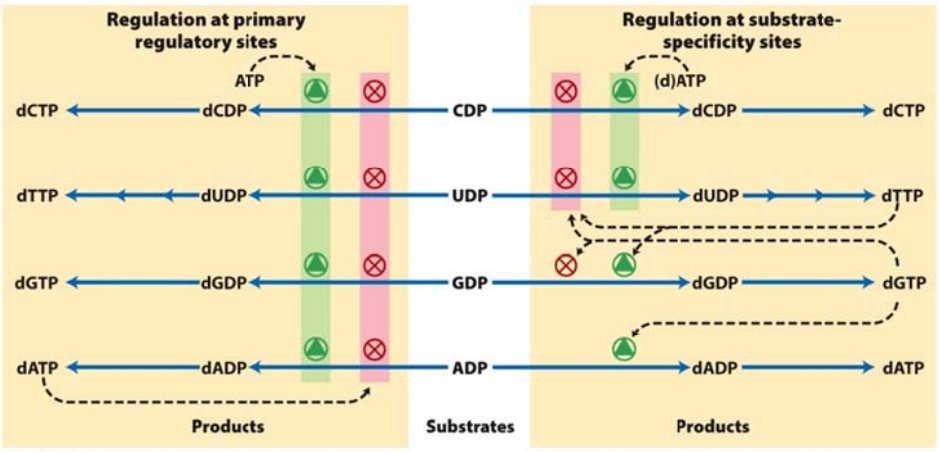


Figure 22-44
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Synthesis of dTMP (Thymidylate)

- DNA contains thymine instead of uracil
- Thymine nucleotides are always synthesized in the form of deoxyribonucleotides
- The precursor is the dUMP
- The synthesis of dTMP from dUMP involves the methylation of uracil and is catalyzed by thymidylate synthase
- - **The donor** of the methyl group is a cofactor containing tetrahydrofolate called **methyltetrahydrofolate**

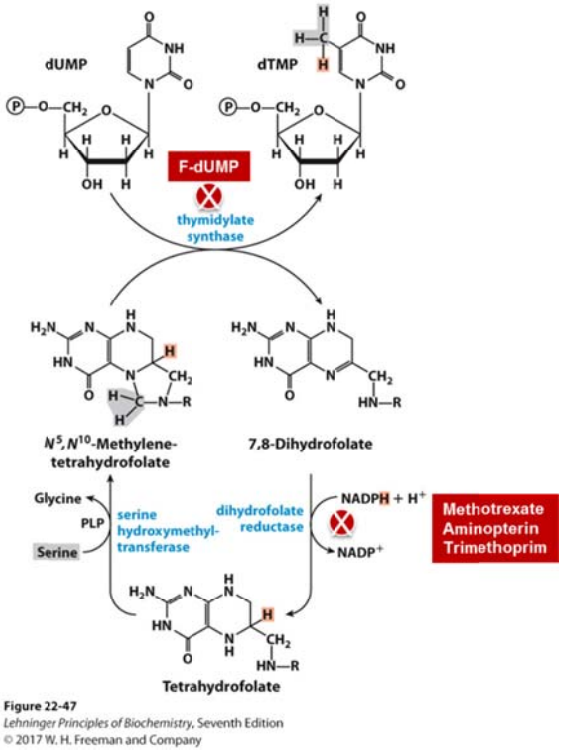


Figure 22-47
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- During the methylation reaction, the cofactor becomes deactivated in the form of dihydrofolate

-It is necessary to reactivate the cofactor for the following reactions of thymidylate synthesis.

- First, dihydrofolate reductase reduces dihydrofolate to tetrahydrofolate
- Then serine hydroxymethyl transferase methylates it to methylenetetrahydrofolate
- **Folic acid deficiency promotes dUTP levels to increase thus it is wrongly incorporated into DNA**.
- DNA reparing enzymes recognize it and, when trying to repair, they generate cuts in the DNA which favors the appearance of mutations and therefore tumors

-**The synthesis of dTTP is an essential requirement for cell division, for this reason tumor cells are highly dependent on this process**

-Taking advantage of this circumstance, **the synthesis of dTTP has become a therapeutic target**

-**To inhibit thymidylate synthase**, fluorouracil has been developed. This compound is activated in cells to FdUMP which acts as an inhibitor

-**Reactivation of dihydrofolate is another target**. **Dihydrofolate reductase inhibitors such as methotrexate and aminopterin have been developed,** both of which are structural analogs of dihydrofolate.

-However, these inhibitors are not selective in tumor cells so they cause great side effects

Degradation of purines and pyrimidines

- Many foods contain large amounts of nucleic acids
- Nucleic acids in the diet resist stomach acid, whereas in the duodenum, pancreatic nucleases and duodenal phosphodiesterases break them down to nucleosides that can be absorbed by the intestinal mucosa
- The fraction of ingested nitrogenous bases that are incorporated into nucleic acids in tissues is very little, since the need for nucleotides is mostly satisfied with de novo synthesis and salvage.

Purines catabolism

- Purines are degraded starting with the release of P by nucleotidases, which gives rise to the corresponding nucleoside
- - In the case of adenosine, deamination to inosine occurs by adenosine deaminase

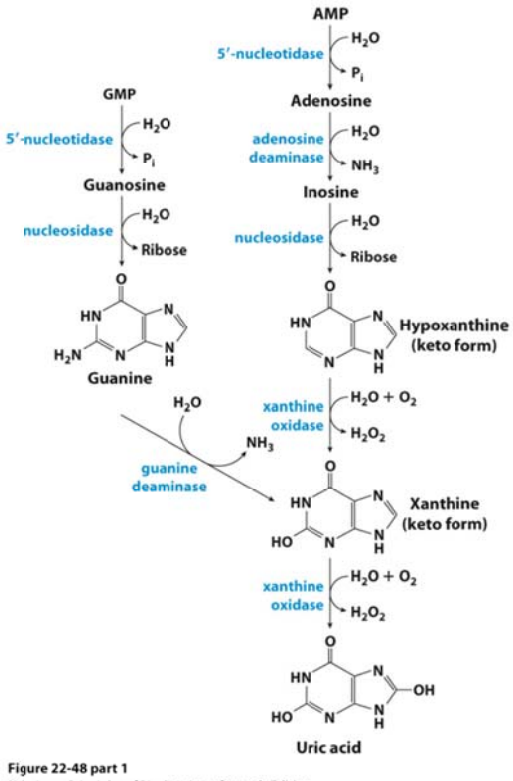


Figure 22-48 part 1
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- Subsequently, inosine is separated from the nitrogen base generating hypoxanthine, xanthine and **uric acid as the final product**, in a series of reactions catalyzed by xanthine oxidase.
- Guanosine is also metabolized to xanthine and uric acid
- - **Uric acid is subsequently excreted via kidneys**
- Deficiencies in purine degradation are the cause of very serious human diseases
- **One of the most common pathologies due to the alteration of this process is gout**
- Gout is characterized by an excess of uric acid in the blood and tissues, which causes its accumulation in the joints and in the kidney, causing inflammation and nephritis.
- The cause of gout is not well known, although it is associated with a diet that is excessively rich in glandular tissues or liver, and also crustaceous.

- The treatment of gout is done through dietary advice and drugs such as allopurinol, which inhibits xanthine oxidase. This prevents uric acid formation and favors xanthine and hypoxanthine, which are more soluble

- The second serious syndrome related to the purine degradation process is **Severe Combined Immunodeficiency Syndrome** (SCID), caused by **adenosine deaminase (ADA) deficiency**.
- The deficiency of ADA causes the accumulation of dATP which is a potent **inhibitor of ribonucleotide reductase.**

- This leads to a deficit of dNTPs and therefore highly dividing cells, such as activated lymphocytes, cannot survive causing immunosuppression